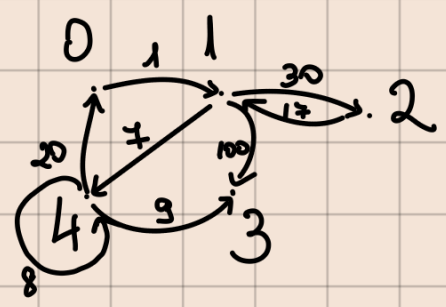
Homework 3



i	j	cost
0	1	1
1	2	30
2	1	17
1	3	100
1	4	7
4	3	9
4	0	20
4	4	8

walk from 0 to 3

path: $0 \to 1 \to 4 \to 3$

cost: 17

Considering 0 as intermediate point:

Distance matrix

$$\begin{pmatrix} 0 & 1 & \infty & \infty & \infty \\ \infty & 0 & 30 & 100 & 7 \\ \infty & 17 & 0 & \infty & \infty \\ \infty & \infty & \infty & 0 & \infty \\ 20 & 21 & \infty & 9 & 0 \end{pmatrix}$$

Previous matrix

$$\begin{pmatrix} 0 & 0 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 & 1 \\ -1 & 2 & 2 & -1 & -1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 0 & -1 & 4 & 4 \end{pmatrix}$$

$k = 1$

D_i

$$\begin{pmatrix} 0 & 1 & 31 & 101 & 8 \\ \infty & 0 & 30 & 100 & 7 \\ \infty & 17 & 0 & 117 & 24 \\ \infty & \infty & \infty & 0 & \infty \\ 20 & 21 & 51 & 9 & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & 1 \\ -1 & 2 & 2 & 1 & 1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 0 & 1 & 4 & 4 \end{pmatrix}$$

$k=2$

D_i

$$\begin{pmatrix} 0 & 1 & 31 & 101 & 8 \\ \infty & 0 & 30 & 100 & 7 \\ \infty & 17 & 0 & 117 & 7 \\ \infty & \infty & \infty & 0 & \infty \\ 20 & 21 & 51 & 9 & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & 1 \\ -1 & 2 & 2 & 1 & 1 \\ -1 & -1 & -1 & 3 & 1 \\ 4 & 0 & 1 & 4 & 4 \end{pmatrix}$$

$k=3$

D_i

$$\begin{pmatrix} 0 & 1 & 31 & 101 & 8 \\ \infty & 0 & 30 & 100 & 7 \\ \infty & 17 & 0 & 117 & 24 \\ \infty & \infty & \infty & 0 & \infty \\ 20 & 21 & 51 & 9 & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & 1 \\ -1 & 2 & 2 & 1 & 1 \\ -1 & -1 & -1 & 3 & 1 \\ 4 & 0 & 1 & 4 & 4 \end{pmatrix}$$

$k=4$

D_i

$$\begin{pmatrix} 0 & 1 & 31 & 17 & 8 \\ 27 & 0 & 30 & 16 & 7 \\ 44 & 17 & 0 & 33 & 24 \\ \infty & \infty & \infty & 0 & \infty \\ 20 & 21 & 51 & 9 & 0 \end{pmatrix}$$

P_i

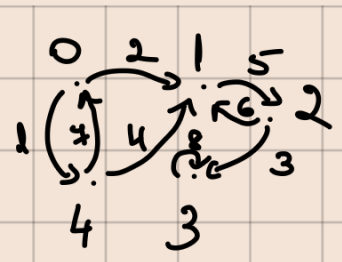
$$\begin{pmatrix} 0 & 0 & 1 & 4 & 1 \\ 4 & 1 & 1 & 4 & 1 \\ 4 & 2 & 2 & 4 & 1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 0 & 1 & 4 & 4 \end{pmatrix}$$

$|V|=5 \Longrightarrow$ stop

The minimum distance from $s=0$ to $t=3$

$P_5(0,3)=4 \quad P_5(0,4)=1 \quad P_5(0,1)=0 = s$

the minimum cost walk: $0 \xrightarrow{1} 1 \xrightarrow{7} 4 \xrightarrow{9} 3$



i	j	cost
0	1	2
0	4	1
4	0	7
4	1	4
1	2	5
2	1	6
2	3	3
3	3	8

Consider vertex 0 as intermediate

D_i

$$\begin{pmatrix} 0 & 2 & \infty & \infty & 1 \\ \infty & 0 & 5 & \infty & \infty \\ \infty & 6 & 0 & 3 & \infty \\ \infty & \infty & \infty & 0 & \infty \\ 7 & 4 & \infty & \infty & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & -1 & -1 & 0 \\ -1 & 1 & 1 & -1 & -1 \\ -1 & 2 & 2 & 2 & 1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 4 & -1 & -1 & 4 \end{pmatrix}$$

$k=1$

D_i

$$\begin{pmatrix} 0 & 2 & 7 & \infty & 1 \\ \infty & 0 & 5 & \infty & \infty \\ \infty & 6 & 0 & 3 & \infty \\ \infty & \infty & \infty & 0 & \infty \\ 7 & 4 & 9 & \infty & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & 1 & -1 & 0 \\ -1 & 1 & 1 & -1 & -1 \\ -1 & 2 & 2 & 2 & -1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 4 & 1 & -1 & 4 \end{pmatrix}$$

$k=2$ D_i

$$\begin{pmatrix} 0 & 2 & 7 & 10 & 1 \\ \infty & 0 & 5 & 8 & \infty \\ \infty & 6 & 0 & 3 & \infty \\ \infty & \infty & \infty & 0 & \infty \\ 7 & 4 & 9 & 12 & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & 1 & 2 & 0 \\ -1 & 1 & 1 & 2 & -1 \\ -1 & 2 & 2 & 2 & -1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 4 & 1 & 2 & 4 \end{pmatrix}$$

$k=3$

D_i

$$\begin{pmatrix} 0 & 2 & 7 & 10 & 1 \\ \infty & 0 & 5 & 8 & \infty \\ \infty & 6 & 0 & 3 & \infty \\ \infty & \infty & \infty & 0 & \infty \\ 7 & 4 & 9 & 12 & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & 1 & 2 & 0 \\ -1 & 1 & 1 & 2 & -1 \\ -1 & 2 & 2 & 2 & -1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 4 & 1 & 2 & 4 \end{pmatrix}$$

$k=4$

D_i

$$\begin{pmatrix} 0 & 2 & 7 & 10 & 1 \\ \infty & 0 & 5 & 8 & \infty \\ \infty & 6 & 0 & 3 & \infty \\ \infty & \infty & \infty & 0 & \infty \\ 7 & 4 & 9 & 12 & 0 \end{pmatrix}$$

P_i

$$\begin{pmatrix} 0 & 0 & 1 & 2 & 0 \\ -1 & 1 & 1 & 2 & -1 \\ -1 & 2 & 2 & 2 & -1 \\ -1 & -1 & -1 & 3 & -1 \\ 4 & 4 & 1 & 2 & 4 \end{pmatrix}$$

$D_5(3,1) = \infty \Rightarrow$ there is no walk from 3 to 1